delay between sending and receiving information. 4G suffers from a 100 milliseconds (ms) latency, while 5G reaction time can be as low as 1 ms, which makes everything happen virtually instantly.

The other advantage is the dense connections that 5G offers. In a given square kilometre, 5G will be able to connect 10 times as many devices as 4G. For this, 5G uses a new radio technology in a different bandwidth and hooks onto portable miniature base stations that run on minimal power to operate and can be densely placed to prevent signals from dropping. Over time, 5G can help connect self-driven cars, so that they become aware of every other car, bike and pedestrian around them; smart cities based on 5G could make anything electric also connected and aware; it might also, in future, help devices at home connect directly to a wireless internet service provider rather than use a WiFi router.

ADVANTAGE ENTERPRISES

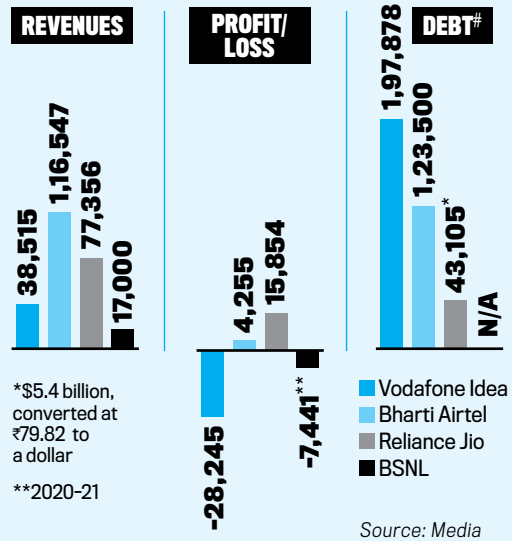
On the enterprise front, one big emerging opportunity for 5G has to do with technology companies looking to expand their ecosystems. These companies spend a fair amount of money on providing end-use applications to customers, but they have only limited connectivity. There are two types of technology companies. First, 'hyperscalers' like Facebook and Google, whose end-customers trawl large platforms. Here, telecom firms provide connectivity as well as SMS service to the end-customer, so that these tech companies can engage with their customers using services like OTPs (one-time passwords). These tech companies are now eyeing a part of the spectrum, using which they will be able to engage directly with their customers and also reduce their telecom costs over some time.

The second type of technology companies are those that want to provide a very concentrated ecosystem for large enterprises. Large corporates that run factories or underground mines require sophisticated infrastructure. Critical to this is what are called 'mesh technologies' or networks that dynamically self-organise and self-configure, reducing installation overheads. These companies pay a huge cost to maintain high-speed infrastructure to transmit the information needed to engage with their customers. Some of them are trying to create a "one-customer ecosystem" with their super app, as the Tata Group is doing with its Tata Neu app. Companies that want to go through a digital transformation have to embed some of these technologies in their existing networks, but connectivity is a huge issue. When private networks come in, these corporates want to have dedicated networks for their use. For instance, Google is running public networks for many enterprises, and so are Microsoft and Amazon. Along with the cloud technologies they offer, these tech companies also provide the connectivity needed by the user companies. With 5G coming in, these tech firms will apply for licences so that they can provide access to cloud solutions at a more affordable cost to small and medium enterprises too. This is already happening in countries such as Germany, the US and Sweden. In fact, 5G is now available across upwards of 70 countries. China, for instance, has reported 233 million 5G customers in the first quarter of 2022.

Meanwhile, global players such as Singtel of Singapore and Verizon of the US have started focusing on teleconnectivity for such B2B platforms. One such platform is healthcare where high-speed precision is required, if remote medical services are to be truly enabled. In the US, even remote surgery is being experimented with now. Work on how precision manufacturing can be done using 5G is under way in these countries. The transportation system and logistics now rely a lot on IoT (internet of things) sensors. 5G enables this because it can handle several times more devices in a given square metre area. In India, many hospital providers want to go virtual, which requires better connectivity. "Enterprises will now have the opportunity to move to the public and private cloud. That shift from on-premises to the cloud will be accelerated," says an official with a Mumbai-based telecom firm.

A HIGH-STAKES GAME

HOW HEALTHY ARE TELCO FINANCES (Figures for 2021-22, in Rs crore)



*$5.4 billion, converted at ₹79.82 to a dollar

**2020-21

As on March 2022

Source: Media reports/ company releases